YOLO26n for Infrared Small-UAV Detection: Baseline and PConv Experiments

Yaorong Huang, Chunshen Yu, Yifan Zhai, and Zhiwen Luo
Macao Polytechnic University

***Abstract*—We study YOLO26n for infrared UAV bounding-box detection on SIRST-UAVB. Exact-pixel duplicates and conflicting labels were audited before fixing a background-view holdout of 1,438 training and 505 validation images. The baseline completed 92 full epochs and part of epoch 93; its selected checkpoint achieved 56.2% mAP@0.5 and 23.2% mAP@0.5:0.95. A layer-1 pinwheel-shaped convolution (PConv) variant completed 92 full epochs under the same 300-epoch learning-rate plan and achieved 35.8% and 14.1%, respectively. Both runs had finite training losses, detection metrics, and saved weights, but some validation-loss entries were NaN. The strict history check stopped the comparison queue before P2 training. We report these observed results and the unresolved numerical issue.**

***Index Terms*—Infrared UAV detection, YOLO26n, small targets, dataset audit, baseline experiments.**

I. Background and Study Design

Small infrared UAVs contain few pixels and limited texture. Bright background structures and birds can resemble the target, while small coordinate errors strongly affect box overlap. We study bounding-box detection using YOLO26n as the lightweight baseline [1], [2]. Pinwheel-shaped convolution (PConv) provides directional shallow sampling for infrared small-target features [3]; a stride-4 P2 prediction branch provides a second way to preserve spatial detail.

The implementation contains B0 (baseline), B1 (a layer-1 PConv replacement), and B2 (an added P2 branch). This report records B0 and a completed 92-epoch B1 attempt. B2 has model-construction and checkpoint-restoration checks but was not trained because B1 failed the subsequent numerical-history check.

II. Dataset Audit and Preparation

SIRST-UAVB contains 3,000 grayscale images at 640 by 512 pixels [3]. The acquired native labels contain 2,961 UAV and 1,747 bird boxes, identified by cross-checking masks and the authors' annotation description. We retain native boxes and use UAV as the training class; bird boxes remain available for error analysis.

The original 2,400/600 split shares 496 exact-pixel image groups; three further groups have conflicting labels. We isolated conflicts, six UAV images without corresponding positive mask pixels and their copies, and close background views at the split boundary. Deduplication and background-view review produced Table I.

The release provides no original video or session identifiers. The grouping therefore represents inferred background-view families, with nearby facade views held out for validation. It has zero cross-split exact-pixel duplicates and no independent test split. The image list, class map, grouping policy, and prepared-manifest digest are frozen in the repository. Conversion preserves original pixels and dimensions; training then uses aspect-ratio-preserving resizing and padding to a 640-pixel input.

TABLE I
Fixed SIRST-UAVB Background-View Holdout

Quantity	Training	Validation
Images	1,438	505
Native UAV boxes	1,421	491
Bird boxes retained for analysis	925	227

TABLE II
Effective Baseline Training Settings

Setting	Value
Input / batch size	640 pixels / 4
Optimizer / initial learning rate	AdamW / 0.002
Weight decay	0.0005
Seed / workers	42 / 4
Mixed precision	Enabled; AMP check passed
Epoch cap / full epochs completed	300 / 92
Final partial epoch	93
Checkpoint selection	Highest validation mAP@0.5:0.95

III. Model and Experimental Settings

B0 uses the n-scale, single-class YOLO26 architecture with prediction strides 8, 16, and 32 and its end-to-end output branch. The constructed training model has 2,504,190 parameters. For consistent initialization across variants, we transfer only same-name, same-shape tensors in layers 0–16 from the official YOLO26n checkpoint: 348 tensors were copied. Layers from 17 onward are randomly initialized. Thus the experiment uses a shared-prefix transfer policy rather than full-model pretrained initialization.

Both runs used one RTX 2080 Ti, Python 3.10.8, PyTorch 2.1.2+cu118, and Ultralytics 8.4.184. B0's automatic optimizer resolved to AdamW. B1 explicitly matches its learning rate 0.002, momentum 0.9, weight decay 0.0005, lrf 0.01, nbs 64, and warmup bias learning rate 0.0. B1 replaces layer 1 with PConv and transfers 342 tensors, excluding that layer. It stops after complete epoch 92 while retaining the 300-epoch learning-rate and augmentation plan. Both use mosaic, scaling, translation, flipping, and the same unaugmented validation split.

0000–0000/00$00.00 © 2021 IEEE

TABLE III
FINAL VALIDATION ON 505 IMAGES AND 491 UAV BOXES (%)

Model state	Precision	Recall	AP50	AP50:95
Partial epoch 93	72.4	48.3	52.6	20.6
B0 best (epoch 81)	66.5	49.2	56.2	23.2
B1 PConv best (epoch 68)	41.6	37.8	35.8	14.1

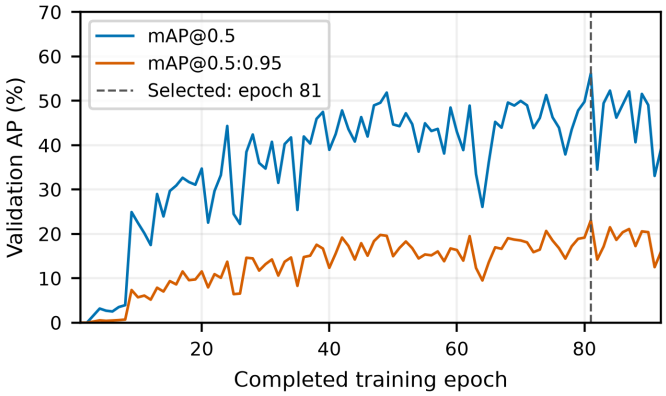


Fig. 1. B0 in-training validation AP from the verified CSV snapshot, through complete epoch 92. The dashed line marks the selected checkpoint at epoch 81. The partial final epoch and post-training best-checkpoint revalidation are reported separately in Table III.

A time callback reserved saving and validation time for B0. Although its log reported 93 epochs, the final visible progress was 317/360 batches in epoch 93: we count 92 full epochs plus a partial 93rd. B1 completed exactly 92 full epochs. B0 selected epoch 81; B1 selected epoch 68. No local model training was performed.

IV. RESULTS AND ANALYSIS

Table III reports final checkpoint revalidation, rounded to the console's three-decimal precision. Precision and recall use the validator's operating point; AP50:95 averages IoU thresholds 0.50–0.95. Revalidation after optimizer stripping differs slightly from in-training values: B0's selected row has AP50/AP50:95 of 56.176%/23.061%, and B1's has 35.578%/14.111%.

The history audit found NaN validation losses in seven B0 rows (epochs 6, 7, 12, 31, 32, 36, 37) and eleven B1 rows (3, 6–12, 14, 16, 17). Training losses, detection metrics, and saved floating-point weights were finite in both runs. B1 training returned normally, but the subsequent all-history finiteness check rejected it and stopped the queue before B2. The cause remains unresolved; no rows were removed or imputed.

PConv produced lower observed AP than B0 in this single-seed attempt. Baseline recall of 49.2% leaves missed-target headroom, and the AP50–AP50:95 gap motivates localization analysis. Figure 1 shows considerable epoch-to-epoch variation. The validation-loss issue needs investigation before further algorithm comparisons.

A. Timing and Artifact Verification

The final B0 GPU validation reported 0.1 ms preprocessing, 3.1 ms inference, and 0.1 ms postprocessing per image, using validation batches of eight. These are framework timings for offline images. Camera input, video decoding, display, and device transfer require a separate end-to-end benchmark on the deployment device.

Final best/last files are each 5,388,158 bytes for B0 and 5,394,626 bytes for B1. Remote manifests and local SHA256 digests agree. CPU deserialization verified finite floating-point states in all 708 B0 and 720 B1 tensors per file, with PConv retained at B1 layer 1. Stripping resets stored epochs to -1; CSVs, checkpoint metrics, and stopping records identify progress. The 35-file B1 archive was verified before acknowledgement and platform shutdown. Raw histories and weights are preserved.

V. EDGE DEPLOYMENT AND FURTHER EXPERIMENTS

P2 training remains pending. The next step is to isolate the validation-loss numerical anomaly while preserving the recorded settings and raw histories. Subsequent comparisons will measure recall, AP, bird-associated false detections, target-size effects, and inference cost under the same data protocol.

YOLO26n conversion for H10, Raspberry Pi execution, and real infrared-video timing have not yet been measured for this trained baseline. The deployment stage will record the compiler/runtime combination, conversion success, any accuracy change, and end-to-end video latency. The GPU validation timing above is the current measured reference.

VI. CONCLUSION

The audited YOLO26n baseline achieved 56.2% AP50 and 23.2% AP50:95. A matched 92-epoch PConv attempt achieved 35.8% and 14.1%, providing no observed improvement in this run. Final weights were saved and hash-verified, but nonfinite validation-loss history stopped the queue before P2. Resolving that numerical issue precedes the remaining comparison and device deployment.

REFERENCES

[1] G. Jocher, J. Qiu, M. Liu, S. Lyu, F. C. Akyon, and M. E. Kalfaoglu, "Ultralytics YOLO26: Unified real-time end-to-end vision models," arXiv:2606.03748, 2026. [Online]. Available: https://arxiv.org/abs/2606.03748

[2] Ultralytics, "YOLO26 model documentation," 2026, accessed: Oct. 23, 2026. [Online]. Available: https://docs.ultralytics.com/models/yolo26/

[3] J. Yang, S. Liu, J. Wu, X. Su, N. Hai, and X. Huang, "Pinwheel-shaped convolution and scale-based dynamic loss for infrared small target detection," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 39, no. 9, 2025, pp. 9202–9210. [Online]. Available: https://ojs.aaai.org/index.php/AAAI/article/view/32996